

Homework

EVOLVING DOCUMENT

Plaksha: Robot motion simulation and optimization

-Andy Ruina

Date of this version: February 3, 2026

[Hyperlinks look like this.](#)

1 Problems

1. **Matlab.** Master all parts of the Pratap Matlab book that we posted on Moodle
2. **Numerical derivative.** Use the definition of derivative to calculate the derivative of $\sin(x)$ at $x = 1$.
 - (a) Compare that to the exact value (exact value is $\cos(1)$). [Practically speaking, ‘exact’ means full computer precision before the compounding of roundoff errors.]
 - (b) Plot error vs h (log log plot) and note that there is a best h because of the competition between Method error and roundoff error.
 - (c) Challenge: Using the Matlab resolution of Matlab’s smallest number “eps” (look it up), estimate the best h for this calculation and compare It with the minimum on your graph.
3. **Tutorial problem.** Complete the exercises from the Tutorial on Thursday morning 29 January.
4. **Euler solver.** Demonstrate that you can write code equivalent to the code shared with you on 30 January. That code separates the solution of an ODE into
 - i) a script (problem set up) and two matlab functions:
 - ii) The Right Hand Side (RHS) function; and
 - iii) The Euler solver (that can solve any set of first order ODEs). If you do that first, you can use it to make the error plots above. You should be able to write this using Matlab help and no more (no Google, no GPT, no copilot).
5. **Make up a problem.** Cook up, or find, some ODE, or set of first order ODEs, and solve them and explore something hopefully-interesting about the solution.

6. Higher order methods.

- (a) Calculate the derivative of $\sin(x)$ using central difference. Make a log log plot of error vs h . On one plot show Euler's method and the central difference method. You should notice that
- the best h is slightly bigger with central difference than Euler
 - for given h the error is much smaller with central difference
 - the slope of the curve for central difference is about 2, twice the slope of the curve for Euler (about 1). This indicates that the error for Euler is proportional to h , and for central difference the error is proportional to h^2 . Hence Central Difference is a so called 'second order' method, whereas with Euler, the slope is one indicating that error is proportional to h so it is called a 'first order' method
- (b) Write a new Matlab function that solves arbitrary ODEs using 'the midpoint method'. This is only a slight modification of the Euler solver. Test it with the differential equation $\dot{x} = x$ with initial condition $x(0) = 1$, evaluated at $t = 1$. On one log log plot show the error as a function of h for both the Euler method and the midpoint method.

7. Spring and mass (2D). One end of a negligible-mass spring (k, L_0) is pinned to the origin, the other to a mass (m). There is gravity (g). Initial Conditions (ICs): The initial position is $\vec{r}_0 = x_0\hat{i} + y_0\hat{j}$, and the initial velocity is $\vec{v}_0 = v_{x0}\hat{i} + v_{y0}\hat{j}$. Motion starts at $t = 0$ and ends at t_{end} .

- (a) Find the Equations of Motion (EoM);
- (b) Assume all parameters and IC's above are given.
- Plot the trajectory of the mass.
 - Animate the trajectory of the mass.
- (c) How many ways can you think of checking the numerical solution? Find as many as you can, and do the check. A list of things to check is started here:
- $k = 0$, all else arbitrary: The motion is parabolic flight (including falling straight down as a special case) [Why? The system is then just ballistics from freshman physics];
 - $x_0 = 0, v_{x0} = 0$, all else arbitrary: The motion stays on the y axis [Why? There is no force in the x direction if the mass is on the y axis. Because the initial velocity has no x component, the mass never leaves the y axis];
 - $g = 0, \vec{v}_0 = \vec{0}$, all else is arbitrary: The motion stays on a radial line. And, if the motion does not cross the origin, the motion is that of a harmonic oscillator (sinusoidal oscillations, check by plotting, say x vs t . [Why? (Advanced) Write the EoM and EoMs in polar coordinates
 $\Rightarrow m\ddot{r} = -k(r - L_0)$
 $\Rightarrow$ the harmonic oscillator equation, $m\ddot{r}^* = -kr^*$, where $r^* \equiv r - L_0$].
 - $L_0 = 0$, all else is arbitrary: ____? _____. [Why? ____? _____.] The question marks indicate the interesting features of the solution and the explanation for those interesting features.
 - etc.*

- vi. *etc.*
- vii. ...